University of Engineering & Technology Peshawar

Computer Programming (C++) Lab

Lab 13

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DATE:	14/05/26
Lab Title:	STRUCTURE IN C++

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1 Objective:

The objective of this lab session is to introduce and apply the following concepts in C++:

- Defining and declaring structures
- Accessing structure members using dot notation
- Initializing structure variables
- Copying structure variables
- Using structures to model real-world entities

2 Introduction to Structures:

A structure is a collection of variables of different data types grouped under a single name. Unlike arrays, where all elements must be of the same type, a structure can hold members of mixed types (int, float, char, etc.). The data items in a structure are called its members.

Key characteristics of structures:

- Members can be of different data types.
- A structure type declaration defines the form of the structure, not memory allocation.
- Memory is allocated only when a structure variable is declared.
- The closing brace in the structure declaration must be followed by a semicolon.
- Structures can be nested within other structures.
- Structure variables can be passed to functions either individually or as a whole.

2.1 Syntax:

```
struct StructureName {              // Define structure template
    dataType member1;               // First member
    dataType member2;               // Second member
    // ...
};

// Example structure definition:
struct part {
    int modelnumber;                // ID number of car
    int partnumber;                 // ID number of car part
    float cost;                     // cost of part
};
```

2.2 Example: One Structure Variable:

```
#include <iostream>  // for input/output operations
using namespace std;
// Declare the part structure
struct part {
    int modelnumber;
    int partnumber;
    float cost;
};
int main() {
    // Declare a structure variable of type part
```

```
    part part1;
    // Initialize structure members using dot notation
    part1.modelnumber = 6244;
    part1.partnumber = 373;
    part1.cost = 217.55F;
    // Display structure members using dot notation
    cout << "Model : " << part1.modelnumber << endl;
    cout << "Part  : " << part1.partnumber << endl;
    cout << "Cost  : " << part1.cost << " $" << endl;
    return 0;
}
```

Output

```
Model : 6244
Part  : 373
Cost  : 217.55 $
```

2.3 Example: Two Structure Variables (Copy):

Structure variables can be directly assigned to one another, which performs a member-wise copy:

```
// Declare and initialize book1 using initializer list
part part1 = {6244, 373, 217.55F};
// Declare a second structure variable
part part2;
// Copy all members from part1 to part2
part2 = part1;
```

Output

```
Model : 6244
Part  : 373
Cost  : 217.55 $
Model : 6244
Part  : 373
Cost  : 217.55 $
```

2.4 Example: Employee Information:

```
#include <iostream>  // for input/output operations
using namespace std;
// Declare the Person structure
struct Person {
    char name[50];
    int age;
    float salary;
};
int main() {
    // Declare a structure variable of type Person
    Person p1;
    // Prompt and read full name (up to 50 characters)
    cout << "Enter Full name: ";
    cin.get(p1.name, 50);
```

```
    // Prompt and read age
    cout << "Enter age: ";
    cin >> p1.age;
    // Prompt and read salary
    cout << "Enter salary: ";
    cin >> p1.salary;
    // Display all information
    cout << "\nDisplaying Information." << endl;
    cout << "Name: " << p1.name << endl;
    cout << "Age: " << p1.age << endl;
    cout << "Salary: " << p1.salary;
    return 0;
}
```

Output

```
Enter Full name: Murad Ali
Enter age: 34
Enter salary: 1200

Displaying Information.
Name: Murad Ali
Age: 34
Salary: 1200
```


3 Lab Tasks :

Task 1: Person Structure:

Declare a structure called Person with members name (string), age (integer), and address (string). Initialize and display using dot notation.

```
#include <iostream>  // for input/output operations
#include <string>    // for string class
using namespace std;
// Declare the Person structure
struct Person {
    string name;
    int age;
    string address;
};
int main() {
    // Declare a structure variable of type Person
    Person person1;
    // Initialize person1 members using dot notation
    person1.name = "Ahmed Khan";
    person1.age = 21;
    person1.address = "House 12, Street 4, Peshawar";
    // Display person1 members using dot notation
    cout << "Name: " << person1.name << endl;
    cout << "Age: " << person1.age << endl;
```

```
    cout << "Address: " << person1.address << endl;
    return 0;
}
```

Output

```
Name: Ahmed Khan
Age: 21
Address: House 12, Street 4, Peshawar
```

Task 2: Student Structure:

Declare a structure called Student with members name (string), age (integer), and grade (char). Initialize and display the members.

```
#include <iostream>  // for input/output operations
#include <string>    // for string class
using namespace std;
// Declare the Student structure
struct Student {
    string name;
    int age;
    char grade;
};
int main() {
    // Declare a structure variable of type Student
    Student student1;
    // Initialize student1 members using dot notation
    student1.name = "Sara Malik";
    student1.age = 20;
    student1.grade = 'A';
    // Display student1 members
    cout << "Student Name: " << student1.name << endl;
    cout << "Age: " << student1.age << endl;
    cout << "Grade: " << student1.grade << endl;
    return 0;
}
```

Output

```
Student Name: Sara Malik
Age: 20
Grade: A
```

Task 3: Book Structure:

Declare a structure called Book with members title (string), author (string), price (float), and pages (integer). Declare two variables book1 and book2, initialize and display both.

```
#include <iostream>  // for input/output operations
#include <string>    // for string class
using namespace std;
// Declare the Book structure
```

```
struct Book {
    string title;
    string author;
    float price;
    int pages;
};
int main() {
    // Declare and initialize book1 using initializer list
    Book book1 = {"C++ Primer", "Stanley Lippman", 45.99f, 975};
    // Declare and initialize book2 using initializer list
    Book book2 = {"The C Programming Language", "K&R", 35.50f, 272};
    // Display book1 information
    cout << "--- Book 1 ---" << endl;
    cout << "Title: " << book1.title << endl;
    cout << "Author: " << book1.author << endl;
    cout << "Price: $" << book1.price << endl;
    cout << "Pages: " << book1.pages << endl;
    // Display book2 information
    cout << "--- Book 2 ---" << endl;
    cout << "Title: " << book2.title << endl;
    cout << "Author: " << book2.author << endl;
    cout << "Price: $" << book2.price << endl;
    cout << "Pages: " << book2.pages << endl;
    return 0;
}
```

Output

```
--- Book 1 ---
Title: C++ Primer
Author: Stanley Lippman
Price: $45.99
Pages: 975
--- Book 2 ---
Title: The C Programming Language
Author: K&R
Price: $35.50
Pages: 272
```